

04. Async Iteration aur Async Generators

Ek line me

Jab data **dheere-dheere** / **delay** se aata ho (network, timer), to `for await..of` aur async generators use karte hain.

Async Iterable (manual tarika)

Normal iterable se farq:

	Normal	Async
Method	<code>Symbol.iterator</code>	<code>Symbol.asyncIterator</code>
<code>next()</code> return	seedha object	Promise jo object deta hai
Loop	<code>for..of</code>	<code>for await..of</code>

```
js
let range = {
  from: 1, to: 5,
  [Symbol.asyncIterator]() {
    return {
      current: this.from, last: this.to,
      async next() {
        await new Promise(r => setTimeout(r, 1000));
        return this.current <= this.last
          ? { done: false, value: this.current++ }
          : { done: true };
      }
    };
  }
};

(async () => {
  for await (let v of range) console.log(v); // 1 sec ke gap se 1..5
})();
```

Spread `...` async iterable par **kaam nahi karta** (wo sync iterator maangta hai).

Async Generator (aasan tarika)

`async function*` likho, andar `await` bhi use kar sakte ho:

js

```
async function* generateSequence(start, end) {
  for (let i = start; i <= end; i++) {
    await new Promise(r => setTimeout(r, 1000));
    yield i;
  }
}

(async () => {
  for await (let v of generateSequence(1, 5)) console.log(v);
})();
```

Yaha `generator.next()` promise return karta hai, isliye `await` chahiye.

Object me bhi use kar sakte ho:

js

```
async *[Symbol.asyncIterator]() { ... }
```

Real-life Example: Paginated Data

GitHub commits jaise API pages me data dete hain. Async generator se saari pagination chhup jaati hai:

js

```
async function* fetchCommits(repo) {
  let url = `https://api.github.com/repos/${repo}/commits`;
  while (url) {
    const response = await fetch(url);
    const body = await response.json();
    // agla page ka URL 'Link' header me hota hai
    let next = response.headers.get('Link').match(/<(.*)>; rel="next"/);
    url = next?.[1];
    for (let commit of body) yield commit;
  }
}

for await (let commit of fetchCommits("user/repo")) { /* use commit */ }
```

Bahar se sirf simple loop dikhta hai, andar page-by-page fetch hota rehta hai.

Yaad rakho

- Delay wala data = async iterator / async generator + `for await...of`.
- Async generator declaration: `async function*`.
- Big files / streams process karne me ye kaafi kaam aata hai.